9.4 The Comparison Test

Theorem (The Direct Comparison Test): Let $\sum a_n$, $\sum c_n$, $\sum d_n$ be series with nonnegative terms. Suppose that for some integer N, we have $d_n \leq a_n \leq c_n$ for all $n > N$.

(a). If $\sum c_n$ converges, then $\sum a_n$ also converges.

(b). If $\sum d_n$ diverges, then $\sum a_n$ also diverges.

Examples:

(a). The series $\sum_{n=1}^{\infty} \frac{5}{5n-1}$ diverges.

(b). The series $\sum_{n=1}^{\infty} \frac{1}{n!}$ converges.

(c). The following series converges.

$$5 + \frac{2}{3} + \frac{1}{7} + 1 + \frac{1}{2+\sqrt{1}} + \frac{1}{4+\sqrt{2}} + \frac{1}{8+\sqrt{3}} + \ldots + \frac{1}{2^n+\sqrt{n}} \ldots$$

The Limit Comparison Test

Theorem (Limit Comparison Test): Suppose that $a_n > 0$ and $b_n > 0$ for all $n \geq N$ (N an integer).

1. If $\lim_{n\to\infty} \frac{a_n}{b_n} = c > 0$, then $\sum a_n$ and $\sum b_n$ both converge or both diverge.

2. If $\lim_{n\to\infty} \frac{a_n}{b_n} = 0$ and $\sum b_n$ converges, then $\sum a_n$ converges.

3. If $\lim_{n\to\infty} \frac{a_n}{b_n} = \infty$ and $\sum b_n$ diverges, then $\sum a_n$ diverges.

Examples:

(a). $\sum_{n=1}^{\infty} \frac{2n+1}{(n+1)^2} =$

(b). $\sum_{n=1}^{\infty} \frac{1}{2^n-1} =$

(c). $\sum_{n=1}^{\infty} \frac{1+n\ln n}{n^2+5} =$

(d). $\sum_{n=1}^{\infty} \frac{\ln n}{n^{3/2}} =$

Extra Examples:

1. The series $\sum_{n=1}^{\infty} \frac{1}{n^2+n}$ is convergent.
2. The series $\sum_{n=1}^{\infty} \frac{1}{n^2-n+1}$ is convergent.
3. The series $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n+1}}$ is divergent.
4. The series $\sum_{n=2}^{\infty} \frac{(\ln n)^3}{n^{1+\epsilon}}$ is convergent for any $\epsilon > 0$.